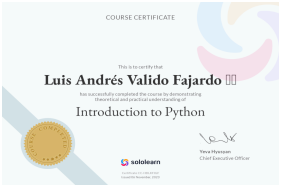
Luis Andrés Valido Fajardo

Education

- **2008-2013** Computer Science Engineering *University of Computer Sciences, Havana*
- **2014** Introduction to the development of the ALBA Comprehensive Supervision and Control System *University of Informatics Sciences,Havana*
- **2014** Introduction to the LaTex text composition system *University of Informatics Sciences,Havana*
- **2014** Science Technology and Society *University of Informatics Sciences,Havana*
- **2014** Creative group work techniques *University of Informatics Sciences,Havana,International Summer School*
- **2015** General Improvement Course for professionals in training *University of Informatics Sciences,Havana*
- **2016-2019** Master in Computer Aided Engineering *University of Matanzas,Matanzas*

Education Online



Thesis

Title: Mechanism for incorporating SVG graphics into the SCADA GALBA HMI.

Tutors: Eng. Roberto Cardenas Isla, Eng. Ariel Guerra Garayta

Description: At the University of Informatics Sciences, Faculty 5 has several production centers among which is the Center for Industrial Informatics (CEDIN) where the Guardian of the ALBA (GALBA), which had the task of providing a solution to the development of a SCADA in conjunction with the PDVSA company.

The Human Machine Interface or HMI (Human Machine Interface) is the module of the GALBA SCADA system that allows configuring, supervising and controlling industrial processes. The HMI provides a library ofgraphic components (CG) that make it possible to recreate as realistically as possible the processes that they occur in plants. These graphics can be simple geometric primitives or as complex as industrial components used in the industry that you want to monitor. The CGs consist of certain

limitation due to its current product design and implementation that are defined by specialists and leaves open to developers its implementation mechanisms and design.

The work was based on the development of a mechanism for incorporating SVG graphics into the HMI of SCADA GALBA with the purpose of solving the limitations present in the graphic components of the HMI. The development of the mechanism is supported by the agile XP methodology with the use of the Scalable Vector Graphics (SVG) standard and Qt framework.

Master's Thesis

Title: ROFLEXIN/LC System Man-Machine Interface Viewer for Mobile Devices.

Tutors: Ph.D Ramón Quiza Sardiñas

Description: The Center for Studies on Advanced and Sustainable Manufacturing (CEFAS) of the University of Matanzas develops the ROFLEXIN/LC project: Robust, flexible and intelligent, low-cost system for monitoring systems and mechanical processes. This project is fundamentally aimed at small and medium-sized companies, where the use of the alternatives available in the world market is not financially feasible. The present investigation arises from the need of the personnel that uses the ROFLEXIN/LC, to visualize the information collected by said system in mobile devices and tablets. The fundamental objective is to develop a mobile application capable of visualizing the information of the ROFLEXIN/LC system in real time. For the development of the proposed solution, an analysis of the main tools, technologies and methodologies used in the construction of software is carried out. The process was guided by the use of the following tools and technologies: {\em Visual Paradigm} as a CASE tool, UML as a modeling language, JSON as a data exchange format, and Java as a programming language. Developed with the Eclipse IDE with the ADT extension. To validate that the results achieved were those expected, a set of tests was carried out. It was possible to correct all the detected non-conformities, which means that the proposed solution is ready to be deployed. Finally, a Viewer for mobile devices and tablets with Android operating system capable of representing in real time the information coming from the ROFLEXIN/LC system was obtained.

Experience

2009-2015 Integral Supervision and Control System of Alba (SCADA GALBA). Developer. *Havana, University of Computer Sciences, Industrial Computer Center*. Module Developer Man-Machine (HMI) belonging to the SCADA Galba project.

2014-2016 Industrial Automated System based on GNU/LINUX (SCADA SAINUX). Developer. *Havana, University of Computer Sciences, Industrial Computer Center*. Module developer Module-Machine (HMI) belonging to the SCADA SAINUX project.

2016-2023 Department of Resources for Learning. Developer, Web Master. *Matanzas, University of Matanzas*. Extensions developer to virtual learning environments of the University of Matanzas (Moodle), educational applications for devices with Android operating system. Administrator of virtual learning environments of the University of Matanzas. Installation and deployment of websites with the use of CMS and LMS.

2016-2022 Informachy Department. Associate Professor. *Matanzas, University of Matanzas*. Teaching category: Instructor

2021-2022 Combiomed Digital Medical Technology. Developer. *Havana*. Module developer Module-Machine (HMI) of the Software for the Control and Monitoring of the Artificial Fan for COVID-19 patients.

2021-2026 Vocational Preuniversity Institute of Exact Sciences *Carlos Marx* of Matanzas. Competitive programming coach. *Matanzas*

2023-2024 Network Department. Specialist B. *Matanzas, University of Matanzas*. Technical support of the Virtual Learning Environments of the University of Matanzas. Installation and deployment of websites using LMS Moodle.

2023-2024 INNOVAT. Developer. *Matanzas*. Design and implementation of the PSGAME+ application for Android mobile devices.

2023-2024 Merchise Autrement. Developer. *Business Solutions on Odoo, consulting and implementation*. Implementation of functionalities. Corrections of non-conformities.

2024-2026 Desoft. Developer. Business Solutions on Odoo. Implementation of addons.

2025-2025 Idoo Group. Developer. *Mobile app development | Mobile app design Tijuana | App development Tijuana | Web design Tijuana - IdooGroup*. Implementation of functionalities. Correction of non-conformities. Implementation of add-ons and systems

Teaching

2016-2016 Python programming language, Informatics Engineering. *Matanzas, University of Matanzas.* 5th year

2016-2022 Computer architecture, Informatics Engineering. *Matanzas, University of Matanzas.* 2nd year

2017-2020 Specific preparation for ICPC competitions, Informatics Engineering. *Matanzas, University of Matanzas.* 1st, 2nd and 3rd year

2019-2019 Android course, Bachelor of Computer Education. *Matanzas, University of Matanzas.* 3rd year

2019-2022 Introduction to programming, Informatics Engineering. *Matanzas, University of Matanzas.* 1st year

2020-2022 Design and object-oriented programming, Informatics Engineering. *Matanzas, University of Matanzas.* 1st year

Thesis tutoring

2015 *System for the Information Management of Technical Files at the Industrial Computer Center*. Yadriel Cuesta Hechavarría. Informatic Science University, Havana, Diploma work to opt for the title of Computer Science Engineer

2015 *Network State processor for SCADA-GALBA adjustments to the electricity sector*. Carlos Manuel Castillo Chacón, Jose Manuel Acevedo Medina. Informatic Science University, Havana, Diploma work to opt for the title of Computer Science Engineer

2016 *Graphic components for the representation of pipe accessories in the SAINUX SCADA*. Rosmery Pedraza Ceballo. Informatic Science University, Havana, Diploma work to opt for the title of Computer Science Engineer

2016 *Graphic components for the representation of valves and pumps in the SAINUX SCADA*. Claudia María González Fernández. Informatic Science University, Havana, Diploma work to opt for the title of

Computer Science Engineer

2016 *Desktop customer for the display environment of the Men's Machine SAINUX SCADA*. Alik Ramón del Risco del Risco. Informatic Science University, Havana, Diploma work to opt for the title of Computer Science Engineer

2016 *Graphic components for the representation of switches and selectors at the SAINUX SCADA*. Javier Bravo Calzado. Informatic Science University, Havana, Diploma work to opt for the title of Computer Science Engineer

2016 *Development of storage graphic components at the SAINUX SCADA*. Luis Orlando Cejas. Informatic Science University, Havana, Diploma work to opt for the title of Computer Science Engineer

2016 *Serialization mechanism for information managed in the editor SAINUX SCADA*. Alvaro Denis Acosta Quesada. Informatic Science University, Havana, Diploma work to opt for the title of Computer Science Engineer

2022 *Plugin to measure the quality indicators of Moodle courses*. Hamsel Brea García. University of Matanzas, Matanzas, Diploma work to opt for the title of Computer Engineering

2022 *Computer System for Human Resources Management and Services of the Artistic Creation Project Vigía*. Ramsey Ricardo Busto Martínez. University of Matanzas, Matanzas, Diploma work to opt for the title of Computer Engineering

2023 *Antencional test system for the “Antonio Guiteras” thermoelectric*. Ronald Guillén Sintes. University of Matanzas, Matanzas, Diploma work to opt for the title of Computer Engineering

2023 *Computer system for physical test management in the sports selection process in the province of Matanzas*. Javier Alejandro González Muñiz. University of Matanzas, Matanzas, Diploma work to opt for the title of Computer Engineering

2023 *Computer system for the commercial value of the tourist offer of the destination city Matanzas*. Randy Mederos Santana. University of Matanzas, Matanzas, Diploma work to opt for the title of Computer Engineering

2023 *Notification plugin of request to create courses in the Moodle to the category managers*. Ana Mónica Rodríguez García. University of Matanzas, Matanzas, Diploma work to opt for the title of Computer Engineering

Thesis opponent

2017 *Software for the management of internal client information in hotel systems from the endonarcoeting perspective*. Fabianne Díaz Santiago. University of Matanzas, Matanzas, Diploma work to opt for the title of Computer Engineering

2019 *Computer Information System for the Directorate of the University of Matanzas, SID*. Rasiel Valdespino Jiménez. University of Matanzas, Matanzas, Diploma work to opt for the title of Computer Engineering

Idioms

Spanish *Native*

English *Professional competence*

Tools and technologies

Programming language

*Extensive experience in developing automation scripts for Linux environments, ranging from Odoo module structure generation and implementation of robust backup systems for platforms such as Moodle and MySQL/PostgreSQL databases, to automation of document processing tasks. Proficient in file and directory manipulation (mkdir, touch, cp, tar), with conditional logic (if) and loops (for, while) to process multiple elements dynamically. Skilled in creating reusable functions with parameter passing, line-by-line file processing, and programmatic generation of content in Python (**manifest**.py) and XML formats. Expertise in designing detailed logging systems that record every execution step, facilitating auditing and troubleshooting. Experienced in managing incremental backups using tar and database dumps with mysqldump (MySQL) and pg_dump (PostgreSQL), including integrity verification and error redirection. Proficient in enforcing automatic retention policies with find. Additionally, skilled in image and document processing tools such as sam2p for JPG to PDF conversion, pdftk for PDF concatenation, and pdfjam for merging documents with custom cover pages. Competent in setting recursive permissions (chmod / umask) and handling command-line arguments with strict validation, ensuring script robustness and reliability in production environments.*

Extensive experience in Python development across multiple domains: customization and creation of modules within the Odoo ecosystem (versions 12 to 18), including PDF/Excel reporting, API consumption, and scheduled tasks; full-stack web development with Django for human resources management, activity planning, tourism resources, and payroll generation in Excel and PDF formats. Implementation of RESTful APIs using Django REST Framework for external application communication, advanced customization of the Django admin interface, and construction of multilingual systems with translatable database content. Proficient in data analysis and scientific computing libraries (Pandas, NumPy, Matplotlib, SciPy, Statsmodels) for building desktop applications focused on multiple linear regression models. Experienced in developing automation scripts for Unix environments (process management, database backups, document manipulation) and solving competitive programming challenges. Capable of designing robust solutions that combine complex business logic, statistical processing, and integration with external services.

Operating systems

Frameworks

Implementation of web systems for the management of human resources, tourist resources, control and planning of activities, generation of payroll for human resources in Excel and PDF format. Implementation of APIs using django-rest-framework for the communication of other applications with a system developed in Django. Customizing the Django admin panel. Implementation of multilingual systems not only in the graphical user interface but also in the content stored in the database.

Content management system

Installation and deployment on servers with Ubuntu and Debian with Mariadb database manager. Personalization, configuration and adinstration of the platform. Administration, management and assembly of courses on the platform. Development of extensions of the report, themes and block types. Development of applications that consume the API that provides the platform. Examples of deployments performed: Eva Undergraduate, Eva Postgraduate, Eva MOOC

ERP

Proven expertise in the full lifecycle of Odoo development, from deployment across versions 12 through 18 to advanced customization and integration. Skilled in modifying existing models and addons as well as building custom modules from scratch. Adept at generating complex PDF and Excel reports, consuming external APIs for seamless data migration, and configuring automated scheduled tasks. Strong front-end capability includes the design of both standard views and complex custom interfaces with integrated graphical components. Experience includes the live deployment of tailored solutions for production use cases at Serena Care and Das-Cortez .

Database manager

2D graphic design

CASE Tool

Web server

Text processor

Version control

Virtualization

I have built multi-container environments using Docker Compose (v3.8) for complex stacks including Odoo 18, PostgreSQL 17, SQL Server 2022, and pgAdmin4. I write custom Dockerfiles that extend official images, adding system dependencies (Python, ODBC, compilers), optimizing layers, and applying security best practices (non-root user, permission and directory management). I configure custom bridge networks for isolated service communication, named volumes for data persistence, and bind mounts for source code, configuration files, initialization scripts, and backups. I manage environment variables, restart policies, and startup ordering. This enables me to quickly spin up portable, realistic development, integration, and data migration environments.

Software development methodology

Extreme Programming

Publications

2017 *Mechanism for the automation of backup copies of the Virtual Learning Environment of the University of Matanzas*. VIII International Scientific Convention of the University of Matanzas, V International Engineering Workshop, Matanzas. Published in citc.umcc.cu with the ISBN code: 978-959-16-3296-8

2017 "*Cesar and the numbers*": *computer resource to promote the learning of the sum operation, in the first grade*. XIX International Event "Mathematics, Statistics and Computing", MATECOMPU 2017, Varadero. Published in citc.umcc.cu with the ISBN code: 978-959-16-3671-3

2018 *Support tool for the probability and statistics subject to graph probability distributions in the distance education study modality of the University of Matanzas*. Opuntia Brava Magazine, Group I Cuban Classification, Las Tunas, Published in the book Science and technological innovation vol II, in the chapter Distance learning. With ISBN for the complete work 978-959-7225-26-3, ISBN for the chapter 978-959-7225-34-8

2018 *Development of the Bezier quadratic and cubic curve graphic components*. Monographs, University of Matanzas ,Matanzas.

2018 *Graphical user interface for interaction with SCADA SAINUX modules (VSERSAI)*. Monographs, University of Matanzas ,Matanzas.

2018 *ROFLEXIN/LC system Man-Machine Interface viewer for Android mobile devices*. Monographs, University of Matanzas ,Matanzas.

2019 *Intelligent and low-cost monitoring architecture for mechanical system and processes*. IX International Scientific Conference, Holguin University, Holguín. ISBN:978-959-7237-34-1

2020 *MiGimusic: APK to promote musical education in pioneering Rhythmic Gymnastics*. International Congress online CUBAMTRICIDAD}{University of Sciences of Physical Culture and Sport "Manuel Fajardo", Havana. ISBN:978-959-07-2395-7

2020 *Technological innovation for musical education in Musical Rhythmic Gymnastics*. 1st National ONLINE Innovation Workshop, National Institute of Sports Physical Education and Recreation, Havana. ISBN:978-959-203-237-8

2021 *The use of mobile devices in Cuban university teaching*. Academia.edu.

2021 *The use of mobile devices. An innovative experience in professional training*. Book: The science of the University of Matanzas in the confrontation with COVID-19. Editorial UM

Events

2008 *Pascal Cup*. University of Computer Sciences,Havana .Contestant

2009 *Pascal Cup*. University of Computer Sciences,Havana .Contestant

2010 *Pascal Cup*. University of Computer Sciences,Havana .Contestant

2011 *Pascal Cup*. University of Computer Sciences,Havana .Contestant

2011 *Cuban Final of ACM-ICPC*. University of Computer Sciences, Havana. Contestant

2011 *My website for Cuba*. University of Computer Sciences,Havana. Relevant

2012 *Pascal Cup*. University of Computer Sciences,Havana .Contestant

2012 *Student Scientific Conference*. University of Computer Sciences, Havana. Three relevant, one outstanding and three mentions.

2012 *My website for Cuba*. University of Computer Sciences,Havana. One outstanding and one mention.

2013 *Student Scientific Conference*. University of Computer Sciences, Havana. Outstanding.

2015 *Pachamama Game Jam*. University of Computer Sciences, Havana. Third place.

2017 *VI UM Programming Cup*. University of Matanzas, Matanzas. Third place, Coach.

2017 *CEPROMEDE Scientific Conference*. Provincial Center of Sports Medicine, Matanzas.

2017 *Cuban Final of ACM-ICPC*. Technological University of Havana José Antonio Echeverría, Havana. Coach

2017 *XIX International Event "The teaching of Mathematics, Statistics and Computing"*. University of Matanzas, Matanzas. Paper: "César and the numbers": Computer resource to favor the learning of the operation sum, in the first grade

2017 *Cyber-physical Systems Seminar: Modeling and Control*. University of Matanzas, Matanzas.

2018 *VII UM Programming Cup*. University of Matanzas, Matanzas. Coach.

2018 *XIV Methodological Scientific Conference of the University of Matanzas*. University of Matanzas, Matanzas.

2018 *Local Constest ACM-ICPC*. University of Matanzas, Matanzas. 2nd place ,Coach.

2018 *ACM-ICPC National Competition*. Military Technical Institute "Jose Martí", Havana. Coach.

2018 *University Forum of Sciences and Techniques*. University of Matanzas, Matanzas. Presentation: Application for monitoring systems and mechanical processes from mobile devices. Mention.

2018 *II International Symposium of the Network of Researchers of Science and Technology: Science and Technological Innovation*. Las Tunas. Presentation: Support tool for the probability and statistics subject to graph probability distributions in the distance education study modality of the University of Matanzas.

2019 *Seminar Computational intelligence for modeling and control of industrial cyber physical system: Research challenges and opportunities*. University of Matanzas, Matanzas.

2019 *VIII UM Programming Cup*. University of Matanzas, Matanzas. Coach and organizer.

2019 *IX International Scientific Convention*. VI International Engineering Workshop, University of Matanzas, Matanzas. ROFLEXIN/LC system man-machine interface viewer for mobile devices.

2019 *IX International Scientific Conference*. Holguin University, Holguín. Intelligent and low-cost monitoring architecture for mechanical system and processes.

2019 *ICPC National Competition*. University of Matanzas, Matanzas. Coach.

2019 *ICPC Caribbean Regional Final*. Eastern University, Santiago de Cuba. Coach.

2020 *First Online Innovation Workshop*. Technological innovation for music education in pioneering rhythmic gymnastics, INDER.

2021 *ICPC Caribbean Finals (Qualifier)*. Provincial Directorate of Education, Matanzas. Preuniversity Teams - Bronze, Coach.

2022 *ICPC Caribbean Finals (Qualifier)*. University of Matanzas, Matanzas. Coach.

2022 *ICPC Caribbean Regional Final*. University of Computer Sciences, Havana. Coach.

2023 *ICPC Caribbean Finals (Qualifier)*. University of Matanzas, Matanzas. Preuniversity Teams - Bronze, Coach.

2023 *ICPC Caribbean Regional Final*. University of Computer Sciences, Havana. Coach.

2024 *Distance education workshop*. Research Institute for Development (IRD), Havana, FSPI-IES Project of the French Embassy, Competitor.

Research

2020 *Educating with José Martí in the Cuban school*. Education Faculty, University of Matanzas. Collaborator

Acknowledgments

2013 *Productive results*. Havana, Informatic Science University, Faculty 5.

2014 *Software developer*. Havana, Informatic Science University, Industrial Computing Center.

2014 *Outstanding*. Havana, Informatic Science University, Industrial Computing Center.

2017 *Guest*. ACM-ICPC Caribbean Regional Final, Villa Clara, Central University of Las Villas "Marta Abreu".

2018 *Guest*. ACM-ICPC Caribbean Regional Final, Havana, Informatic Science University.

2019 *Acknowledgment*. Outstanding Quality Worker, Matanzas, University of Matanzas.

2023 *Acknowledgment*. Competitive programming coach, Matanzas, Vocational Preuniversity Institute of Exact Sciences *Carlos Marx*.

Competitive Programming Coach

IPVCE Regional Cup

Two bronze medals, an equal number of silver medals, and six gold medals.

- Cristhian Ernesto Lugo Dorta
- Zaniel García Orihuela

UCI Programming Cup for Pre-University}

Two mentions, six bronze medals, 12 silver medals, and five gold medals.

- César Raúl Luis González
- Zaniel García Orihuela

LiveCode Cup

One silver medal.

Cuban Computer Science Olympiad}

16 bronze medals, six silver medals, and six gold medals. Fourteen students were promoted to the national pre-selection. 12 students were awarded a university degree directly based on competition results.

- Cristhian Ernesto Lugo Dorta
- César Raúl Luis González
- Zaniel García Orihuela

Ibero-American Olympiad in Informatics

Three mentions, one bronze, and three silver. 6 students participated in this event.

- Cristhian Ernesto Lugo Dorta
- César Raúl Luis González
- Zaniel García Orihuela

International Olympiad in Informatics

One student participated in this event

- Cristhian Ernesto Lugo Dorta

ICPC Qualifier

Three bronze medals and one gold medal in the pre-university team division.

- Cristhian Ernesto Lugo Dorta
- César Raúl Luis González
- Zaniel García Orihuela

ICPC Caribbean Regional Final}

One team qualified for this stage of the event.

- Cristhian Ernesto Lugo Dorta
- Zaniel García Orihuela